

CRISPR-Cas9 Therapeutic Applications in Monogenic Diseases: Clinical Progress, Delivery Challenges, and Ethical Frameworks

A Review of Phase I/II Trials and Regulatory Developments, 2019–2024

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Abstract

CRISPR-Cas9 genome editing has transitioned from a laboratory tool to a clinical intervention with remarkable speed. As of 2024, 47 CRISPR-based therapies are registered in Phase I or Phase II clinical trials, targeting diseases including sickle cell disease (SCD), beta-thalassemia, Duchenne muscular dystrophy (DMD), transthyretin amyloidosis (ATTR), and Leber congenital amaurosis type 10 (LCA10). This review synthesizes published trial results, evaluating efficacy outcomes, adverse event profiles, and delivery modality performance. Exagamglogene autotemcel (exa-cel), the first approved CRISPR therapy (FDA, December 2023), demonstrates that functional cure of hemoglobinopathies is achievable via ex vivo hematopoietic stem cell editing. In vivo delivery, while more amenable to broad application, faces ongoing challenges from immunogenicity, off-target editing, and tissue tropism limitations. We review emerging solutions including lipid nanoparticle (LNP) optimization, base editing, prime editing, and anti-CRISPR proteins for dosage control. We additionally examine the evolving international regulatory landscape and the ethical dimensions of germline editing, post the 2018 He Jiankui controversy.

Keywords: CRISPR-Cas9, gene therapy, sickle cell disease, beta-thalassemia, lipid nanoparticles, genome editing, clinical trials

1. Introduction

1.1 From Discovery to Clinic

The repurposing of the bacterial CRISPR-Cas9 adaptive immune system as a programmable genome editing tool by Doudna, Charpentier, and colleagues earned the Nobel Prize in Chemistry in 2020 and has since generated a therapeutic pipeline with few parallels in the history of medicine. The tool's core mechanism — a guide RNA (gRNA) directing the Cas9 endonuclease to a complementary genomic target where it introduces a double-strand break — is conceptually simple

yet enables an extraordinary range of genetic interventions, from gene disruption and correction to transcriptional regulation and base conversion.

1.2 The Case for Monogenic Diseases

The approximately 7,000 known monogenic diseases collectively affect an estimated 300 million people worldwide. Many involve well-characterized mutations in genes with understood structure-function relationships, making them attractive early targets for genome editing. The 2023 FDA and EMA approval of exa-cel for SCD and beta-thalassemia represents a landmark: not merely a first-in-class approval but proof-of-concept that a single treatment can durably correct the underlying genetic cause of a severe inherited disease.

2. Clinical Trial Landscape

2.1 Hemoglobinopathies

The most clinically mature CRISPR programs target hemoglobinopathies. Exa-cel operates by reactivating fetal hemoglobin (HbF) expression through disruption of the BCL11A erythroid enhancer in autologous hematopoietic stem and progenitor cells (HSPCs). The pivotal trials (CLIMB SCD-121 and CLIMB THAL-111) enrolled 45 and 54 patients respectively. At median follow-up of 29.4 months, 97.1% of SCD patients were free of vaso-occlusive crises and 93.5% of beta-thalassemia patients achieved transfusion independence. Adverse events were primarily attributable to the conditioning regimen (busulfan-based myeloablation) rather than the editing procedure itself.

2.2 Neuromuscular Diseases

DMD trials targeting exon 51 skipping via CRISPR-mediated deletion of exon 50 are in Phase I, with preliminary data suggesting modest but measurable dystrophin restoration in muscle biopsies (mean 1.6% of normal levels, compared to the approximately 3–15% threshold estimated to provide clinical benefit). Delivery via adeno-associated virus serotype 9 (AAV9) has demonstrated acceptable safety in the three cohorts reported to date, though anti-AAV antibody responses in two subjects necessitated dose modifications.

2.3 In Vivo Programs

ATTR amyloidosis has emerged as the leading in vivo CRISPR program. NTLA-2001, an LNP-formulated therapy targeting the TTR gene in hepatocytes, demonstrated dose-dependent reduction of serum TTR protein of up to 93% in Phase I at the highest dose cohort, with a favorable adverse event profile and no off-target editing detected at 48 genomic sites assayed. This program validates the LNP platform for hepatic CRISPR delivery and is now in Phase III.

3. Delivery Challenges and Emerging Solutions

3.1 Immunogenicity

Pre-existing immunity to SpCas9 (the most clinically used Cas9 ortholog), derived from prior *Staphylococcus aureus* infection, is present in 58–79% of adult donors in Western populations.

Neutralizing antibodies can reduce editing efficiency by up to 90% in murine models. Strategies under evaluation include use of alternative Cas9 orthologs (SaCas9, CjCas9), immunosuppressive co-administration, and encapsulation approaches that shield the protein from immune recognition.

3.2 Off-Target Editing

Off-target cleavage at genomic sites with partial complementarity to the gRNA represents a safety concern, particularly for long-lived cells in which oncogenic mutations could propagate. High-fidelity Cas9 variants (eSpCas9, HiFi Cas9) have reduced off-target rates to below the detection threshold of whole-genome sequencing in most settings, though ultra-deep sequencing and cell-based selection assays continue to reveal low-frequency events. Base editing and prime editing, which do not introduce double-strand breaks, offer inherently reduced off-target risk profiles and are advancing into early clinical trials.

4. Regulatory and Ethical Considerations

4.1 International Regulatory Frameworks

The FDA's approval of exa-cel under Biologics License Application reflects a precedent-setting framework for CRISPR therapies as 'gene therapy products' subject to the existing biological products regulatory pathway. The EMA issued a concurrent conditional marketing authorization. The WHO's Expert Advisory Committee on Developing Global Standards for Governance and Oversight of Human Genome Editing published its final framework in 2021, recommending a tiered oversight structure distinguishing somatic editing (permissible with appropriate clinical oversight), heritable germline editing (currently impermissible for clinical use), and enhancement applications (requiring extensive societal deliberation).

4.2 Ethical Dimensions

The unauthorized germline editing of human embryos carried to term by He Jiankui in 2018, resulting in the birth of children with biallelic CCR5 disruption, crystallized concerns about the regulatory oversight gaps that permit rogue applications of powerful technologies. Beyond the immediate harms to the children involved, the case highlighted the inadequacy of voluntary moratorium frameworks and spurred international calls for binding governance instruments. The debate continues regarding the appropriate pathway toward permissible germline editing research, with opinions ranging from permanent prohibition to carefully governed limited research under independent oversight.

5. Conclusions

CRISPR-based therapeutics have achieved clinical validation with the approval of exa-cel, demonstrating functional cure of hemoglobinopathies. The pipeline for additional indications is extensive and the pace of delivery innovation — particularly in LNP formulation and base/prime editing — is rapid. Remaining challenges in immunogenicity management, off-target characterization, manufacturing scalability, and equitable access will determine whether CRISPR therapy benefits are restricted to wealthy health systems or become globally accessible. Robust

international governance frameworks, grounded in both safety evidence and broad societal deliberation, are essential prerequisites for the responsible expansion of this technology's reach.

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